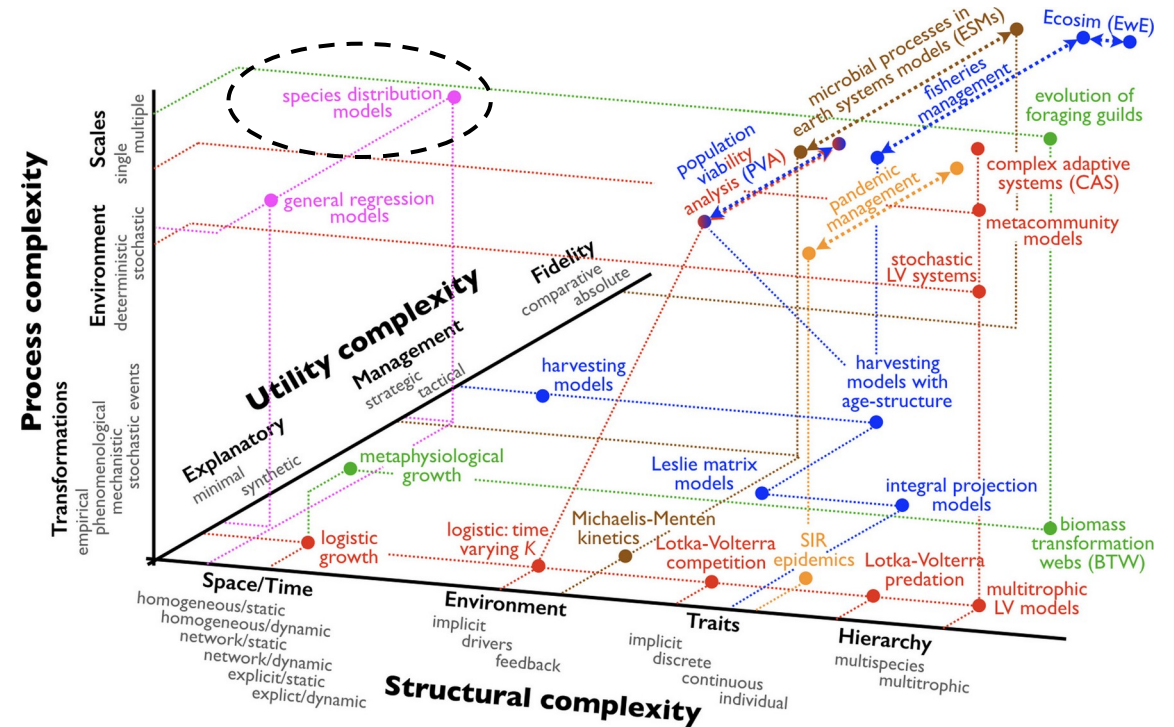
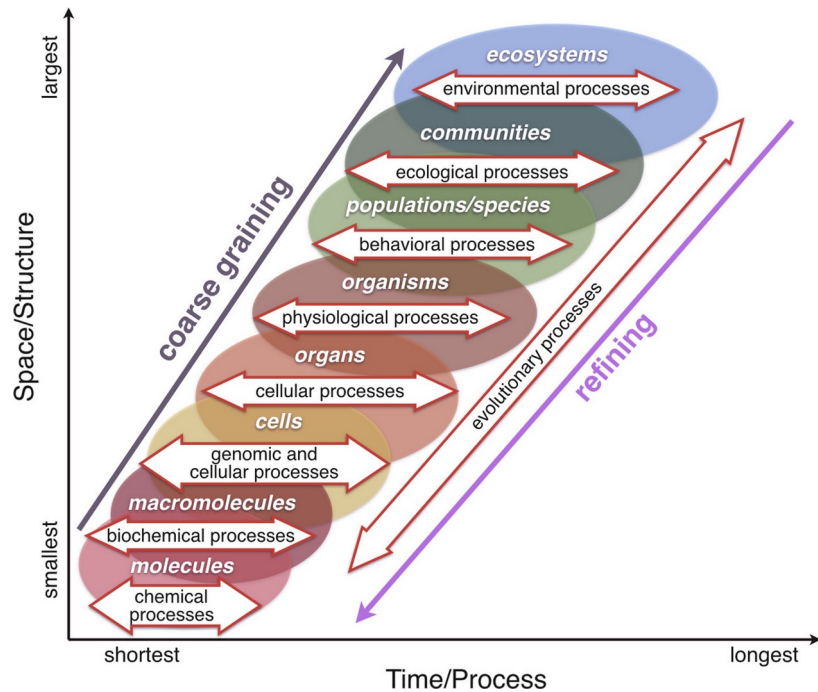


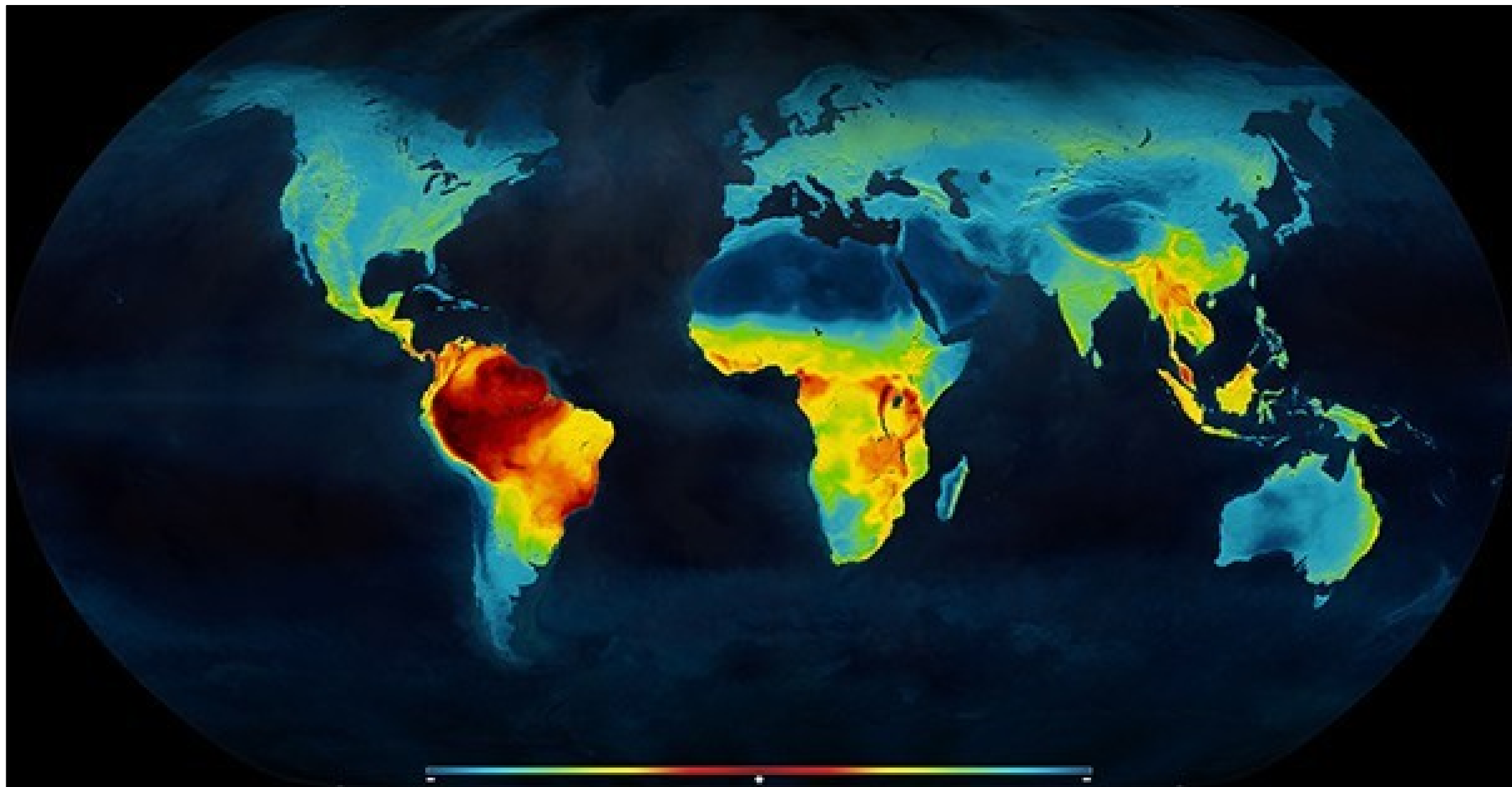
# 生态学和进化生物学中的尺度问题

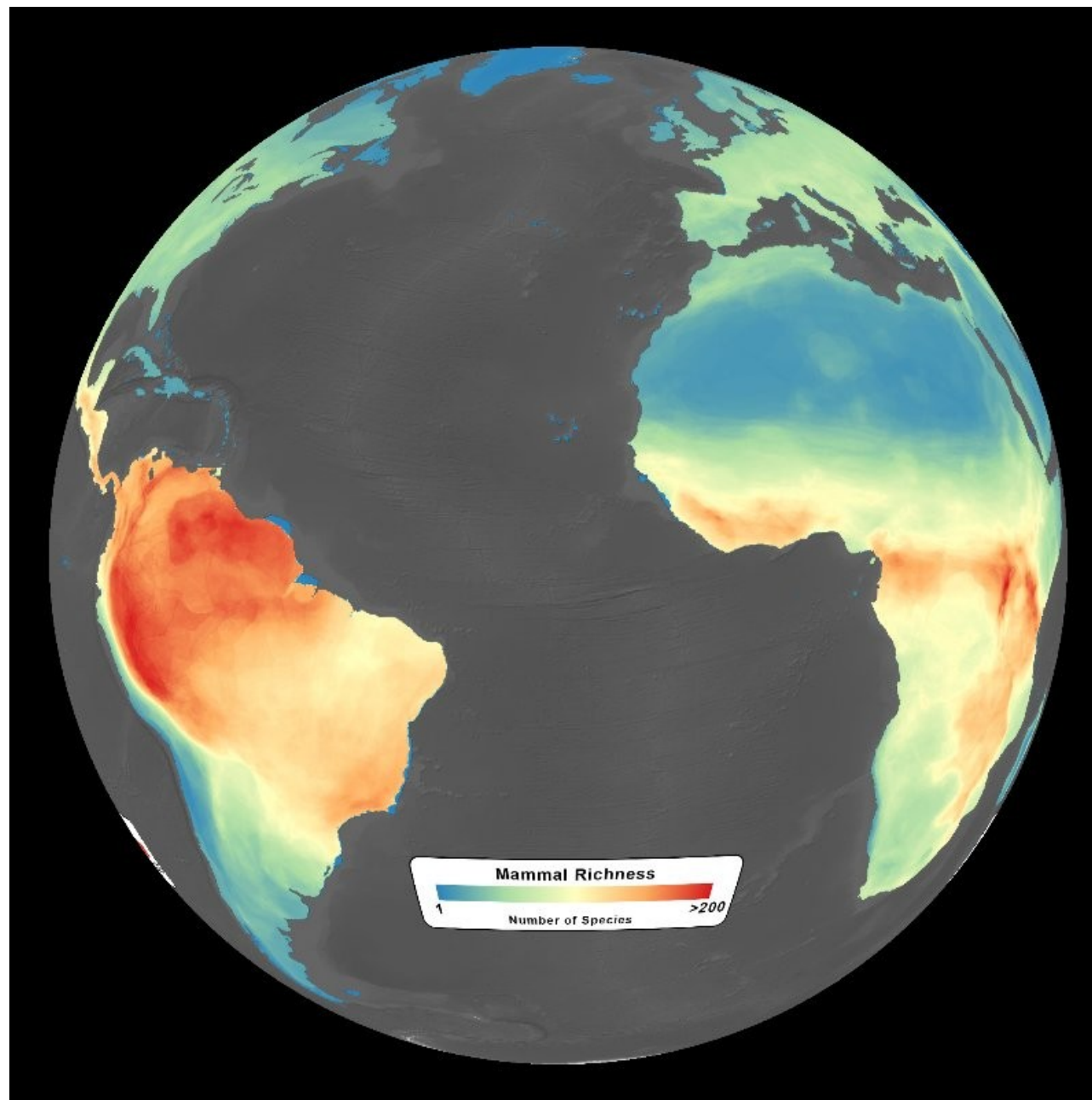
中科院动物所 乔慧捷

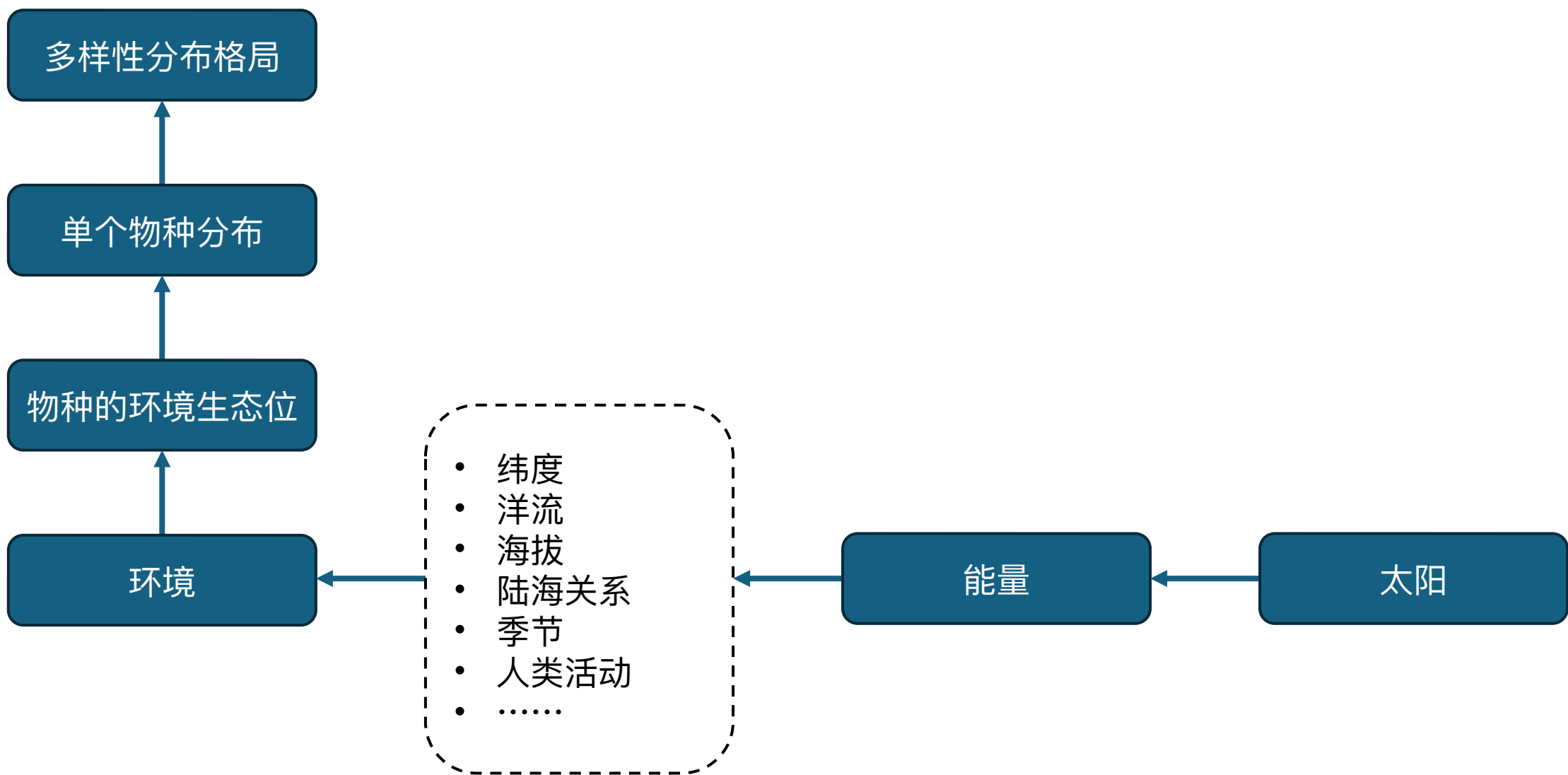
2026-9

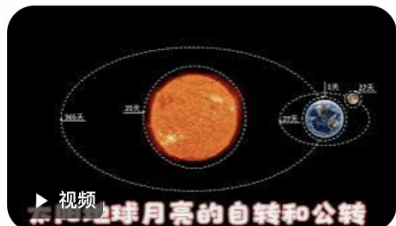
# Ecology is a science of scale





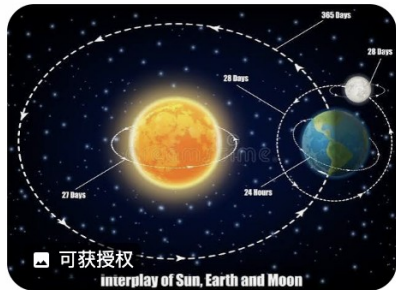






地球和月亮的自转和公转所产生的现象

YouTube



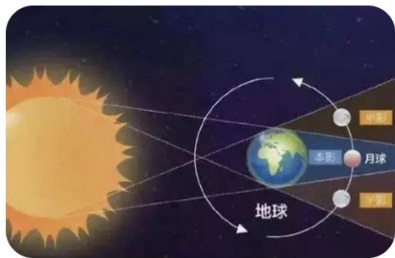
太阳、地球和月亮的相互作用，矢量教...

库存摄影



卡通地球月亮和太阳的排列设计元素素...

六图网



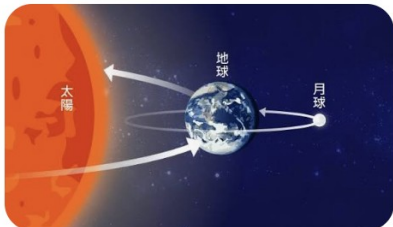
太阳地球月亮三者有什么关系

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为什么地球围绕太阳转，而月亮不围绕...

搜狐



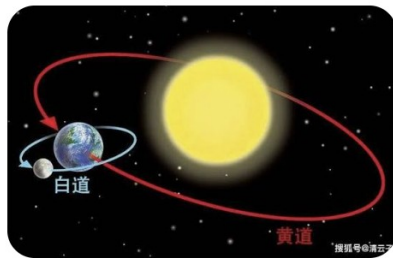
中央氣象署數位科普網-太陽系這一家

中央氣象署數位科普網



太阳地球月亮三者有什么关系

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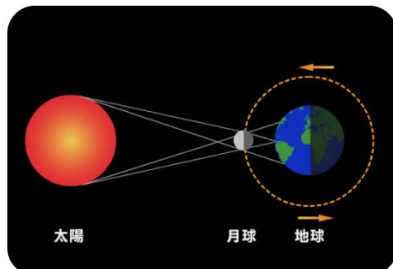
太阳、地球和月亮之间的相互作用是如...

搜狐



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臺北市立天文科學教育館



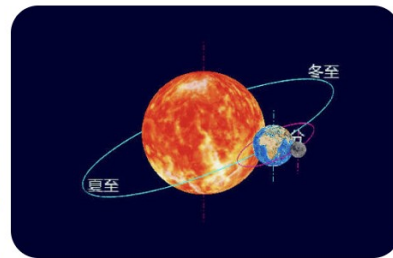
中央氣象署數位科普網-誰把太陽變不見...

中央氣象署氣象會員登入



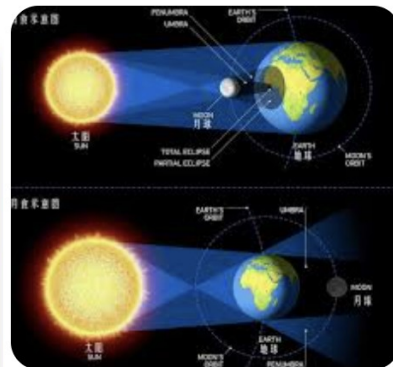
地球、太阳和月亮有什么关系，它们是...

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程序员炫酷溜娃！用代码画地球、日月...

腾讯云- Tencent



当太阳、月球和地球三者连成一条直线...

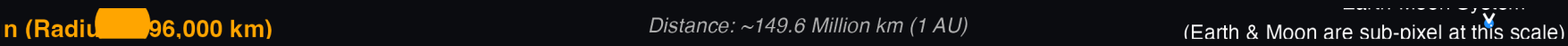
搜狐



Sheep Mandarin Poems 童谣【地球太...

# 1. True Scale (Distances & Sizes Proportional)

At this scale, Earth (~12,742 km) and Moon (~3,474 km) are virtually invisible dots.



# 2. Schematic Diagram (Not to Scale)

Sizes and distances are adjusted for visibility and educational presentation.

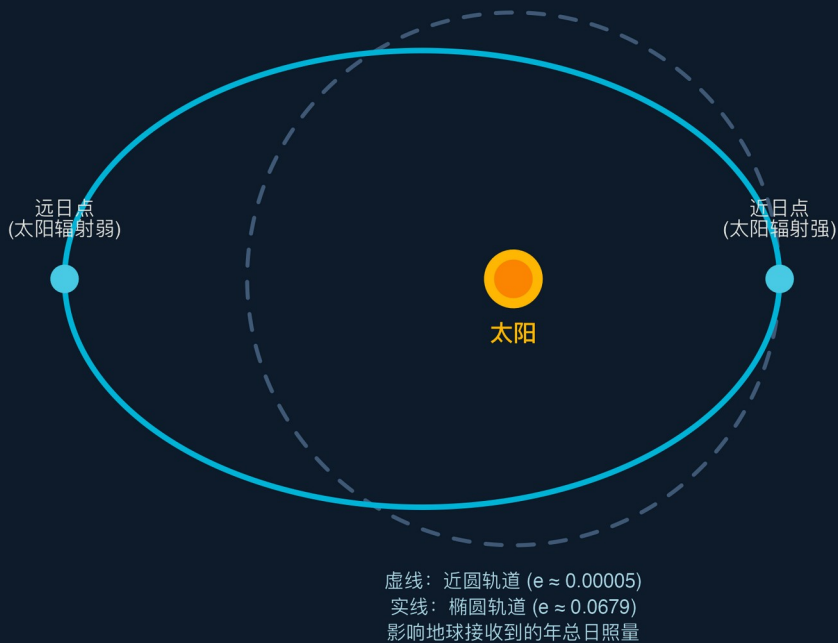


# 米兰科维奇循环三大天文要素示意图

三大轨道参数周期性叠加，驱动了地球地质历史上的冰期与间冰期循环交替

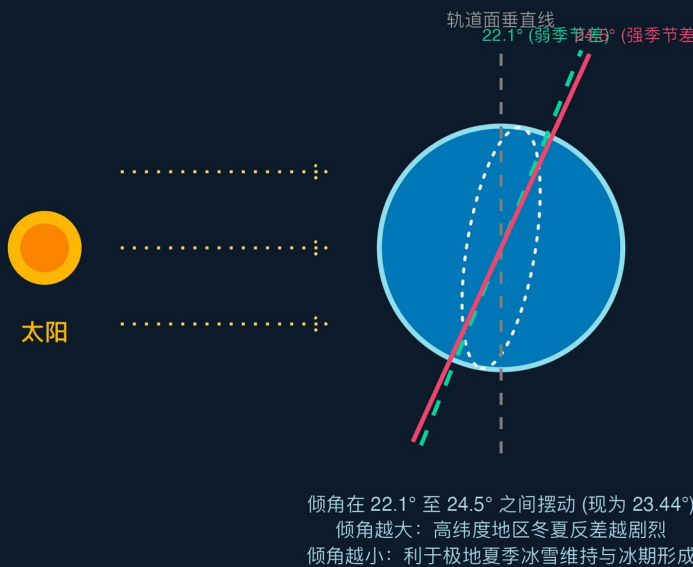
## 1. 偏心率 (Eccentricity)

轨道形状变化 | 周期：约 10 万年



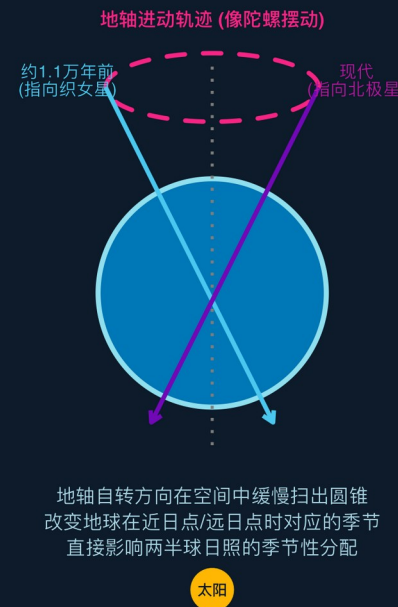
## 2. 地轴倾角 (Obliquity)

地轴倾斜角度 | 周期：约 4.1 万年



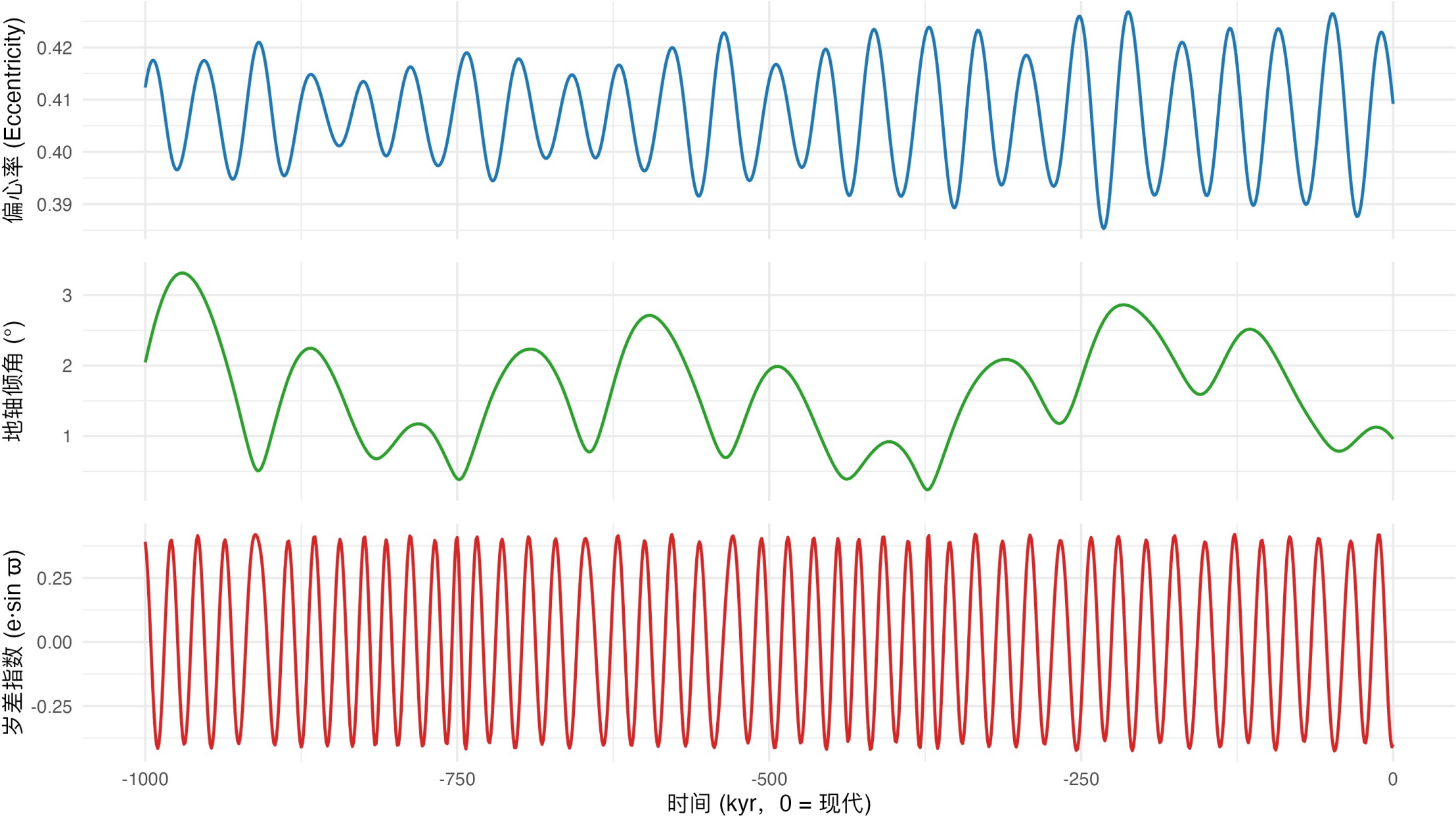
## 3. 地轴岁差 (Precession)

地轴自转进动 | 周期：约 1.9 ~ 2.4 万年



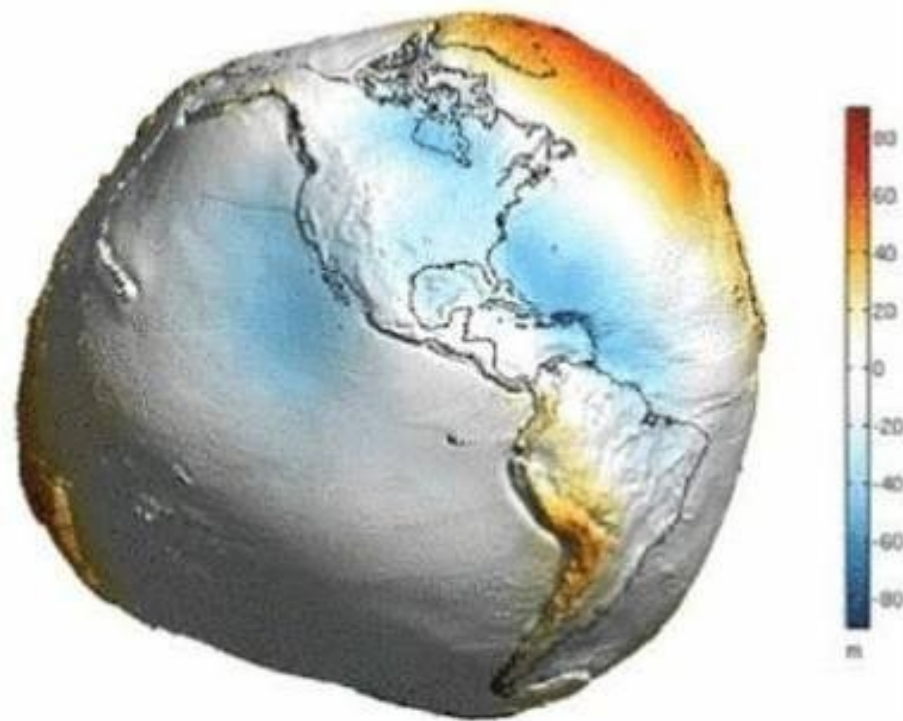
制图：基于 Milankovitch 轨道参数驱动理论

米兰科维奇循环三大参数 (Laskar 2004 解)





如果地球没有水的话，就会变成这样...





公差：40mm ~  
40.6mm  
~1.5%



公差：6378.137 ~  
6356.752  
~0.336%

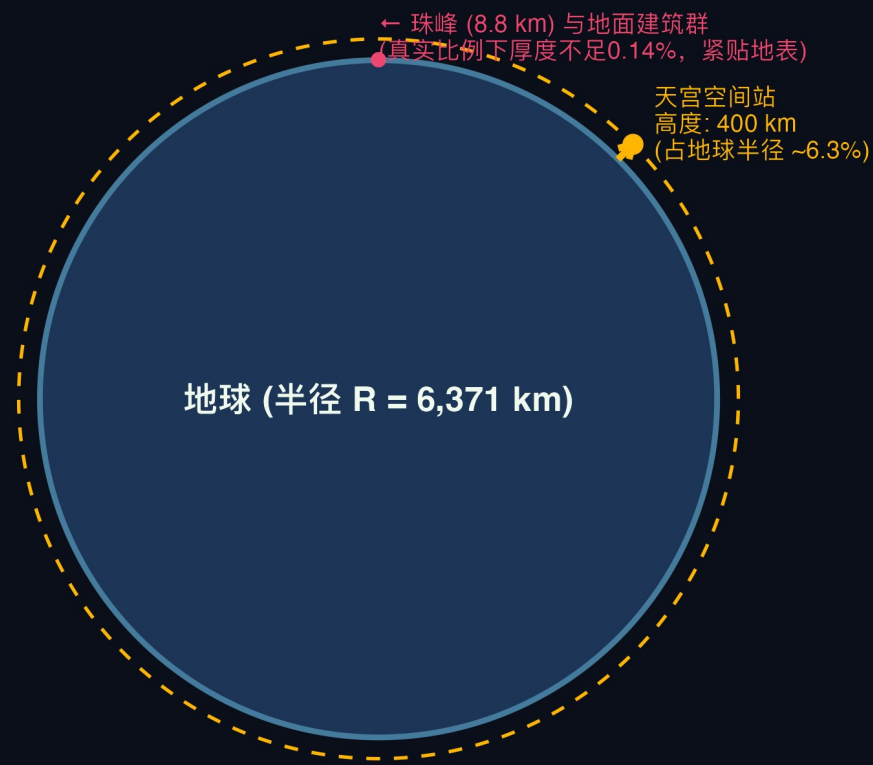


# 地球真实物理尺度与高度关系精确定量图

基于真实地球平均半径 (6,371 km) 与精确海拔高度绘制

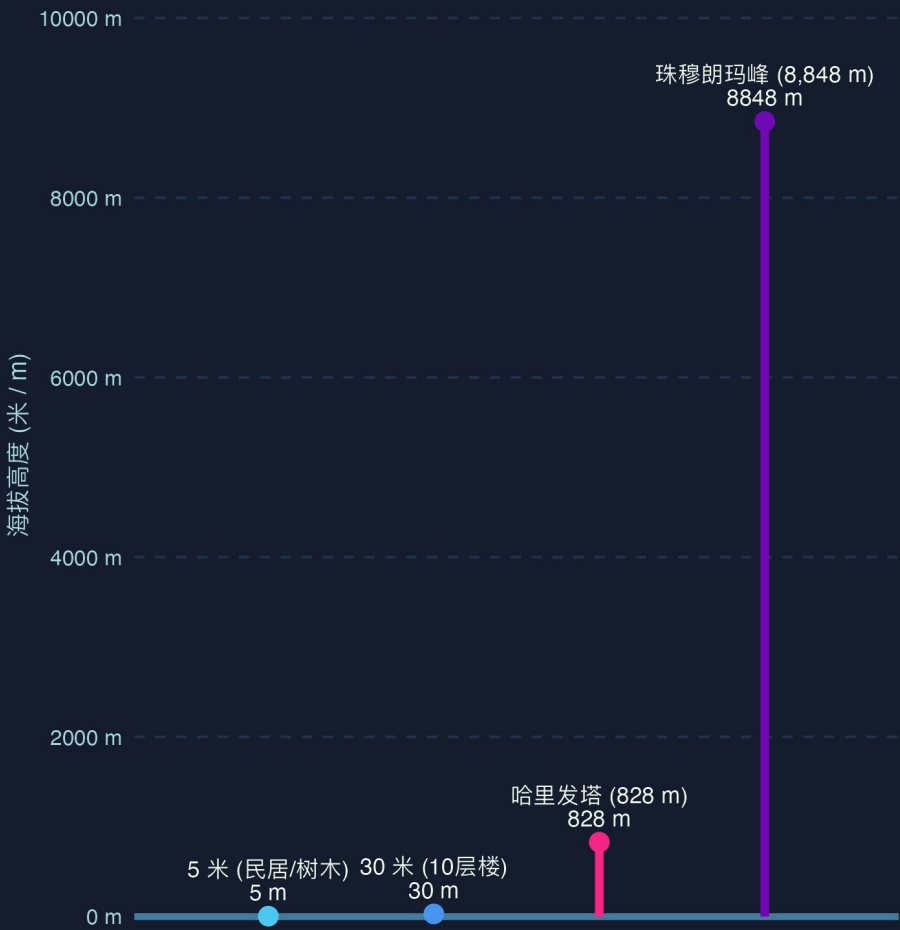
## 全景真实比例视图 (Global Scale)

在整颗地球尺度下，400 km 的空间站清晰可见，而珠峰和建筑完全贴在地表



## 地表局域微观放大 (0 ~ 10,000 米真实比例)

对比 5m、30m、哈里发塔与珠峰的真正物理高差



# MAJOR EVENTS INFLUENCING GLOBAL CLIMATE & SPECIES EXTINCTIONS

## 小行星撞击 ASTEROID IMPACTS

小行星撞击的毁灭性最大损失的伤害。并避免天灾、毁灭破策，它的将火的为破坏道燃的毁灭的毁灭。

- Massive dust cloud 拦截 blocking sunlight
- Global cooling



Impact



Global cooling

66 million years ago  
K-Pg Extinction

## 大陆板块漂移 CONTINENTAL DRIFT

大陆板块漂移隆成，帮助河流故重聚而变流板块漂移，并其他板块漂移，前方非需将及地据狭大据据据据的基地板块等在会间板板的积态。

Pangea

Gondwana & Laurasia

0 Ma



250 Ma



Gondwana & Laurasia

0 Ma



大陆板块漂移  
Dynamic plate tectonics



改变变化  
Changing ocean currents



板套全面化  
Affecting global climate



其他的改变  
Altering habitats

## 大型火山喷发 LARGE-SCALE VOLCANIC ERUPTIONS

大型火山喷发要增大含量： $\text{CO}_2$ 、 $\text{SO}_2$ 、反氧期，每大含量所含的 $\text{CO}_2$ 、 $\text{SO}_2$ 及火冰。

- Ash 灰灰灰
- Global cooling & warming
- Atmospheric changes



Impact



Magma



Environmental change



Permian Extinction events

## 冰河世纪 ICE AGES

冰河世纪毁灭地，低温地地地地地、生流流流流和奇抗的产地性能，毁灭湿地地区死的流流。



Extensive glaciation



Low global temperatures



Sea level changes



Holocene boundary

全球系统

250 Ma

地质年代

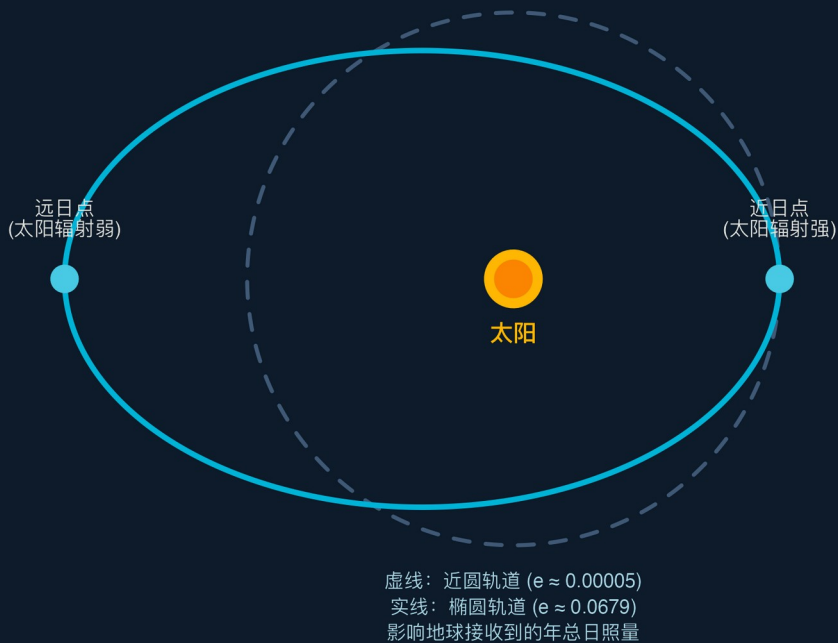
150 Ma

# 米兰科维奇循环三大天文要素示意图

三大轨道参数周期性叠加，驱动了地球地质历史上的冰期与间冰期循环交替

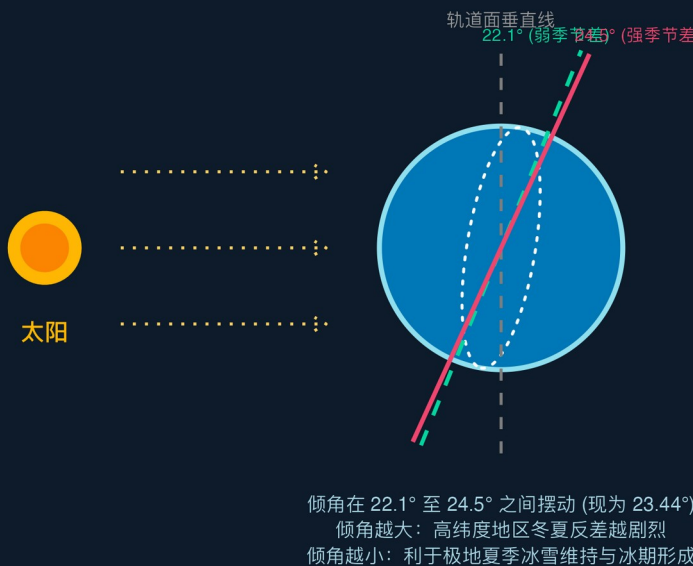
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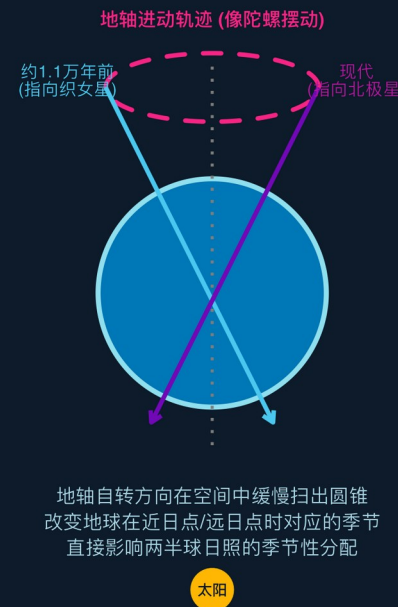
## 2. 地轴倾角 (Obliquity)

地轴倾斜角度 | 周期：约 4.1 万年



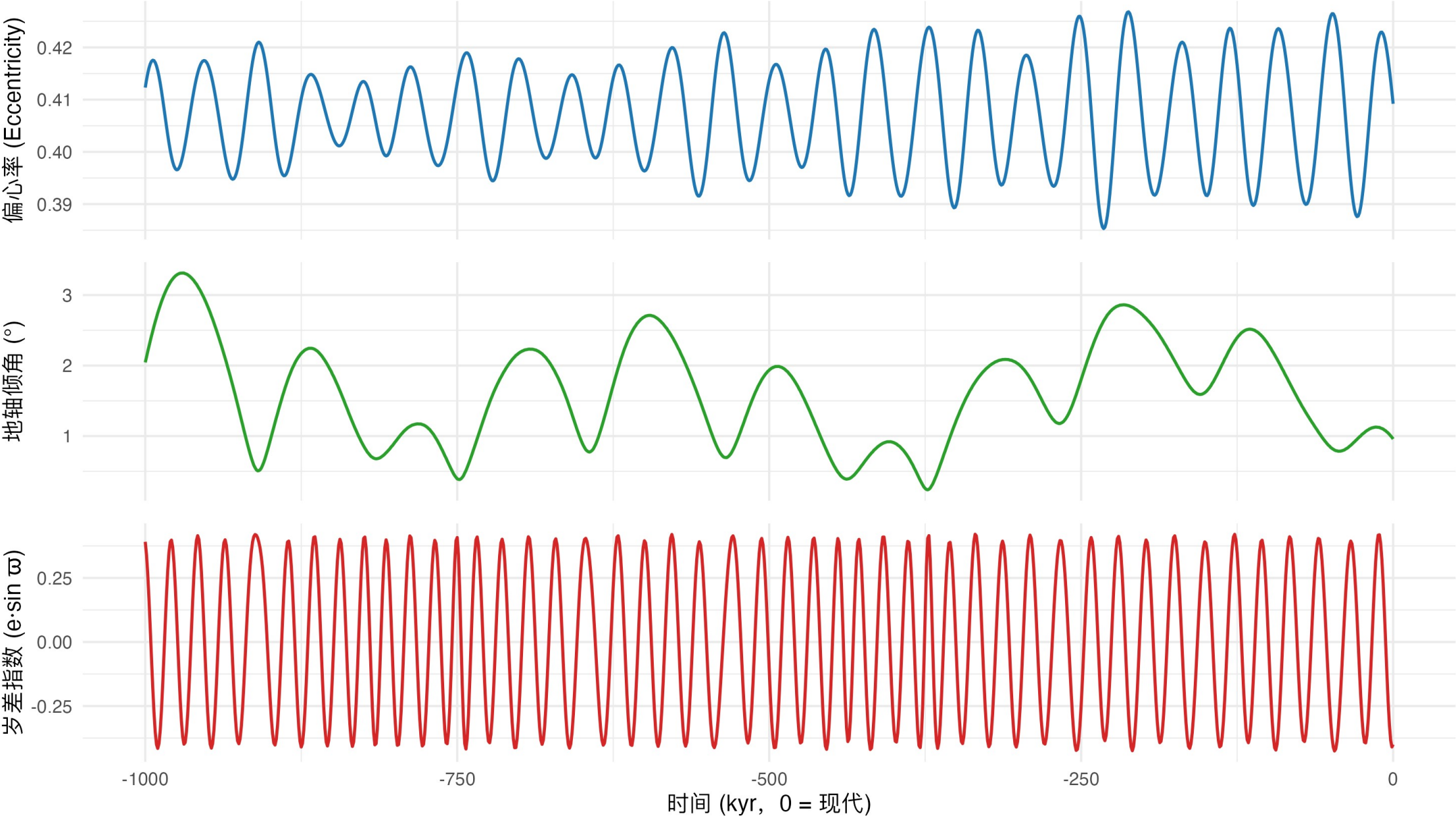
## 3. 地轴岁差 (Precession)

地轴自转进动 | 周期：约 1.9 ~ 2.4 万年



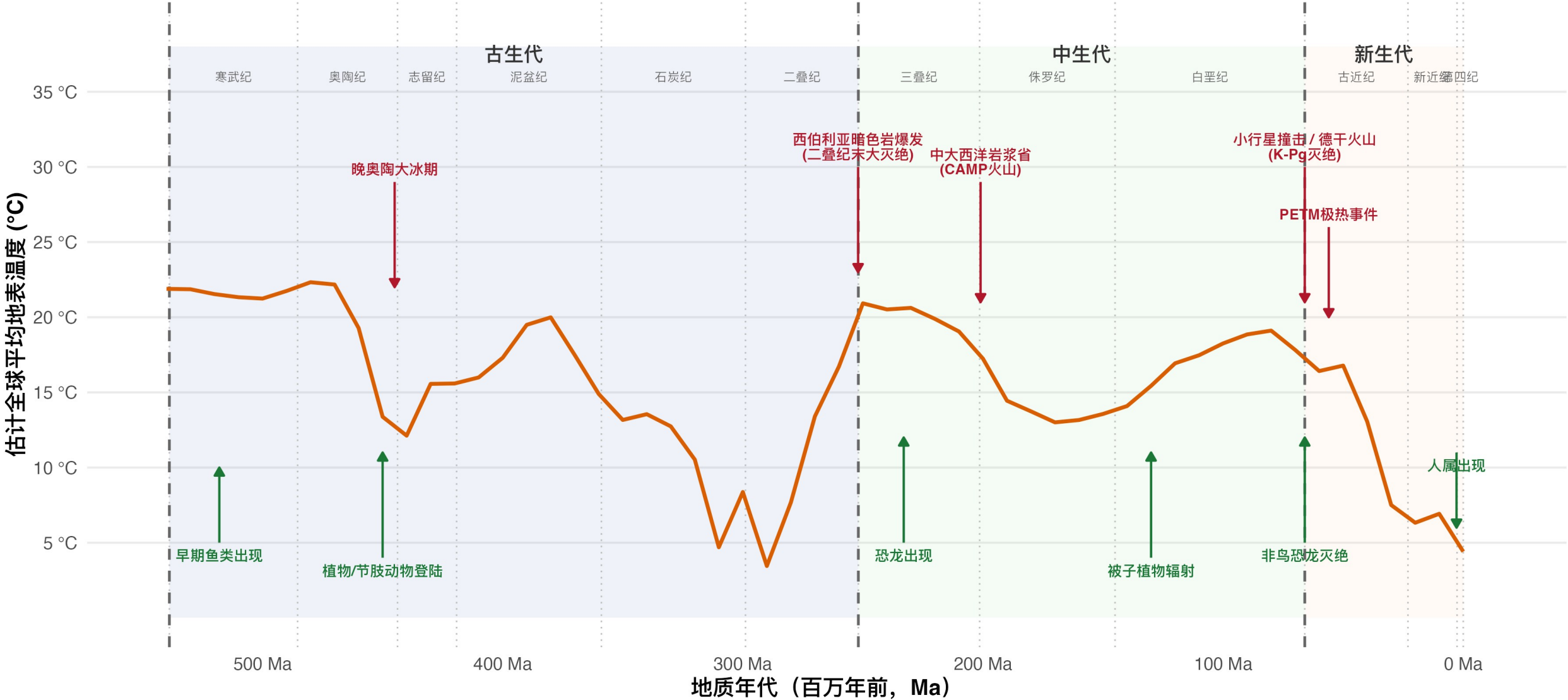
制图：基于 Milankovitch 轨道参数驱动理论

米兰科维奇循环三大参数 (Laskar 2004 解)



# 显生宙（5.4 亿年前至现代）全球平均地表温度与关键历史事件

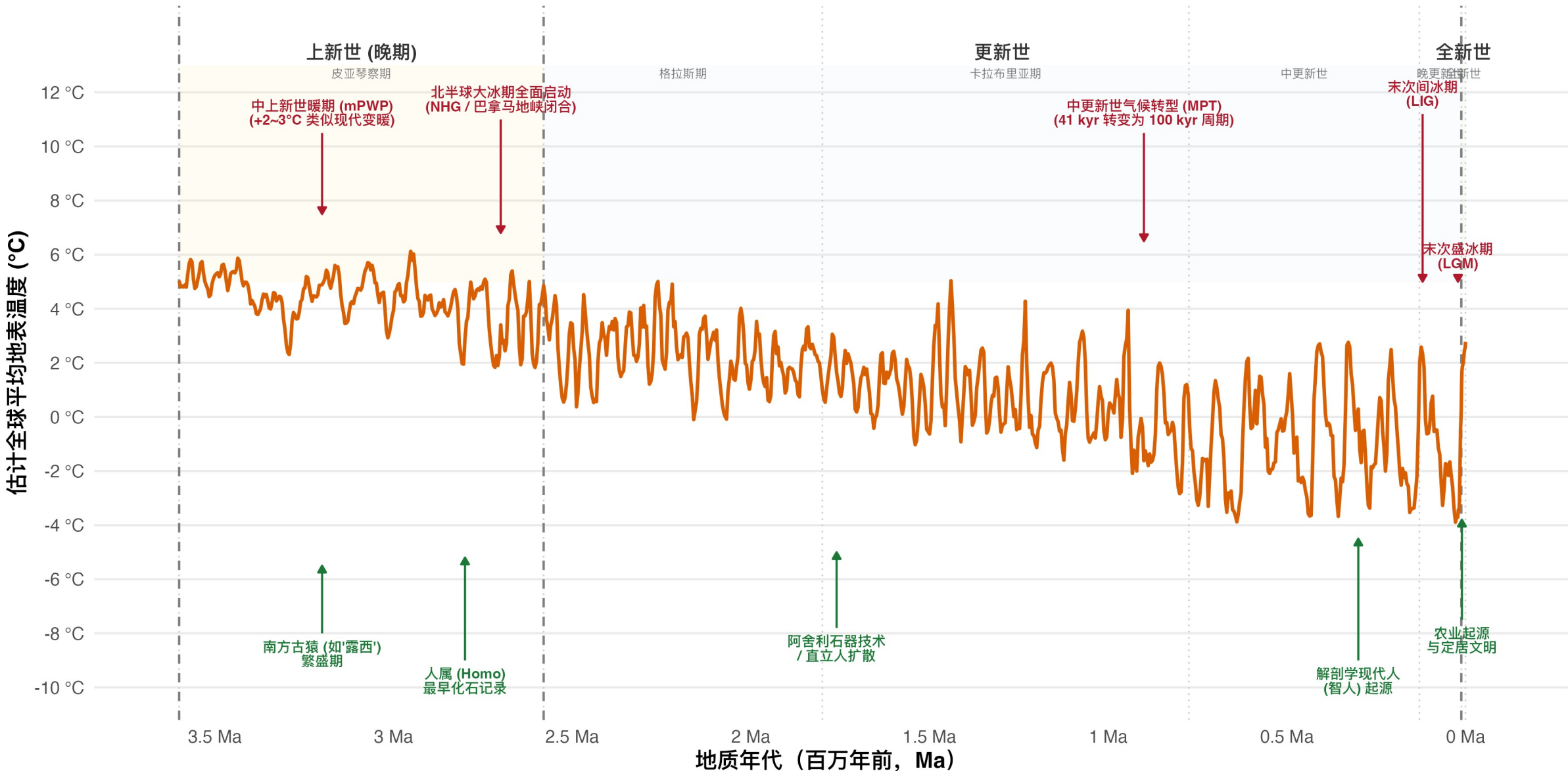
灾变事件/超级火山/撞击 | 底部绿标：重大生物演化事件





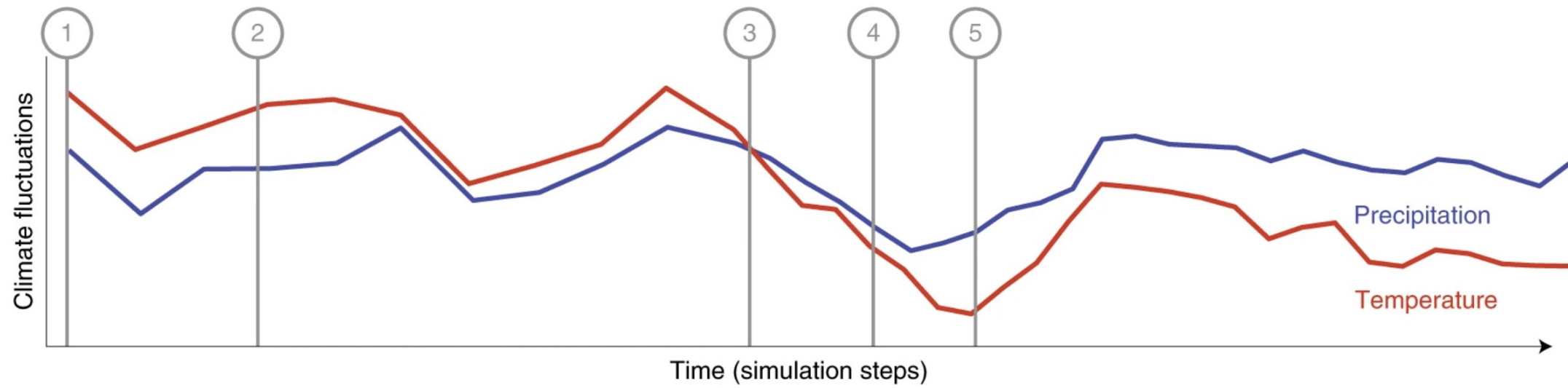
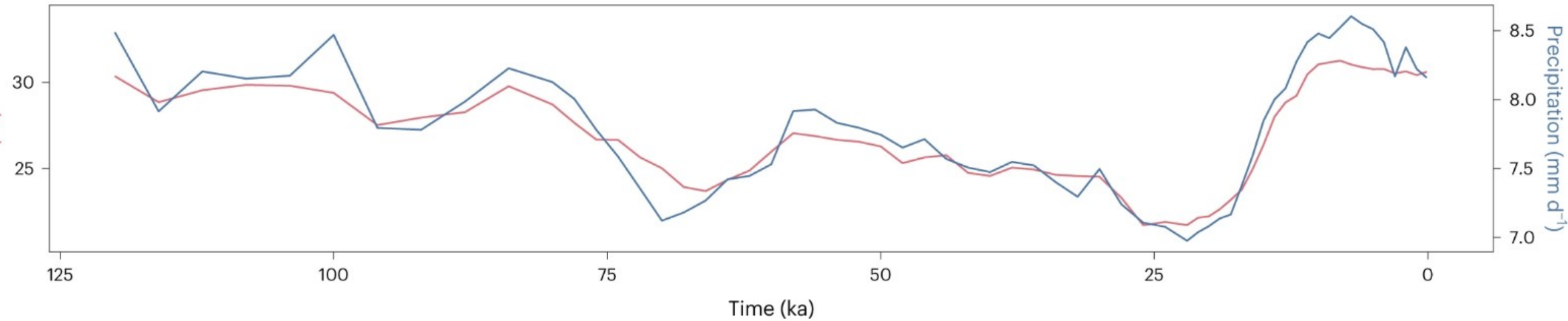
# 晚近地质历史（过去 360 万年）古气候演化与重大地质/演化事件

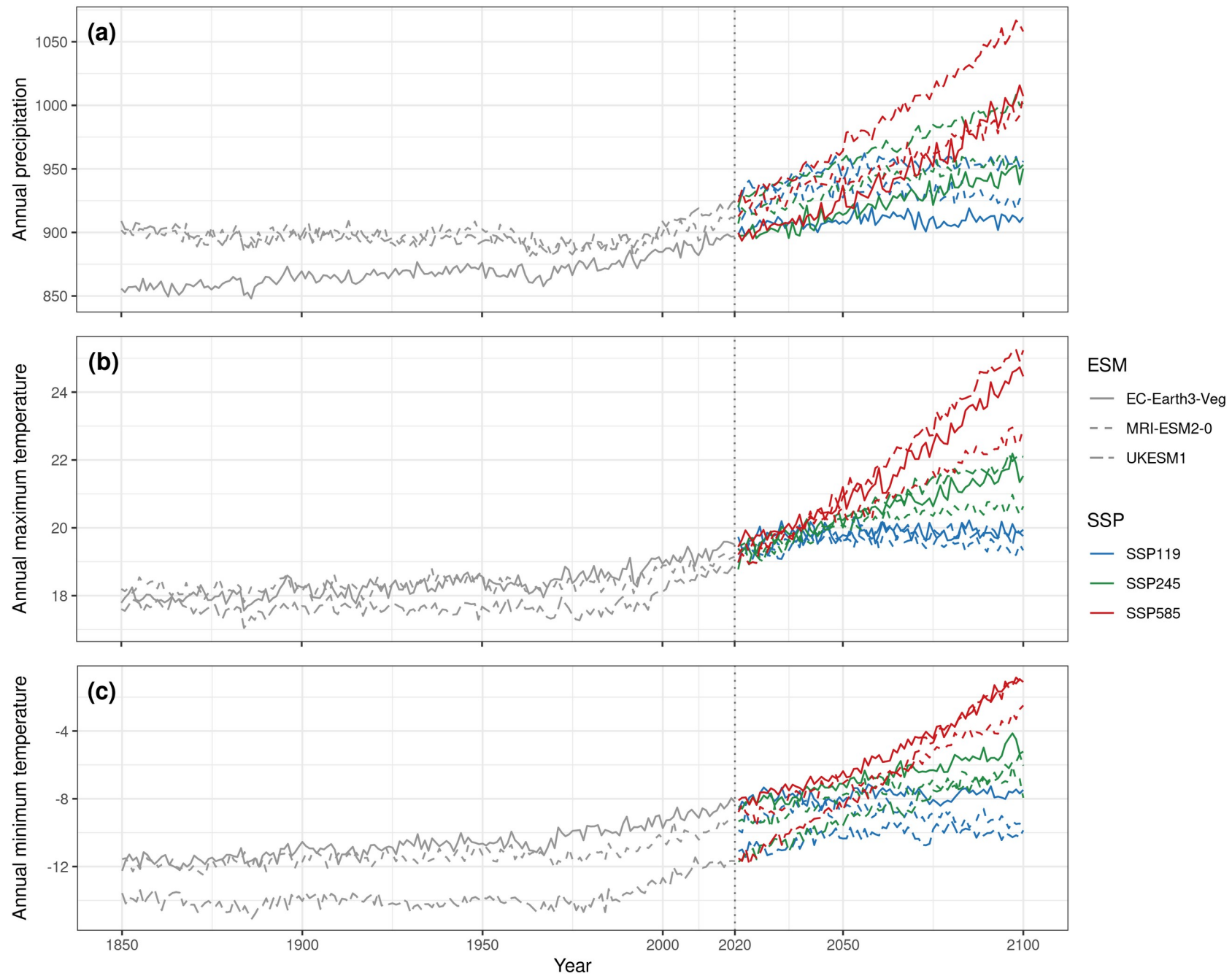
顶部红标：冰期启动与重大古气候事件 | 底部绿标：人类演化与智人物质文化事件



c

Maximum temperature (°C)





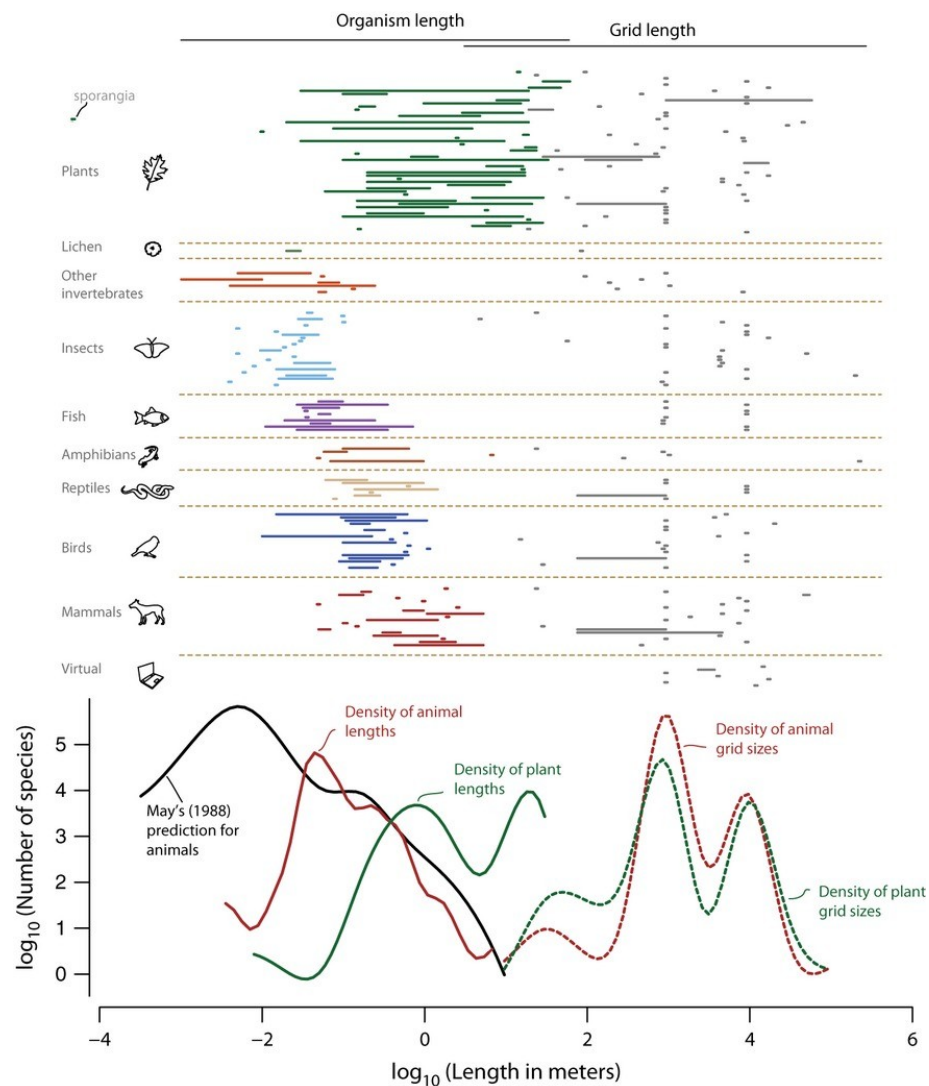
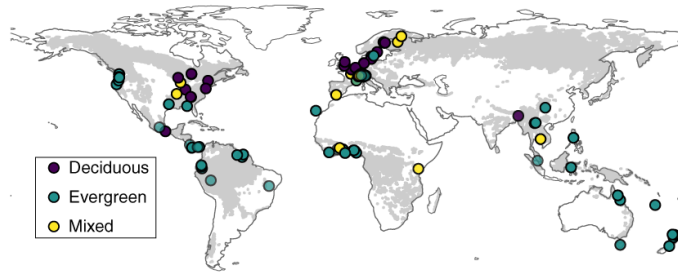
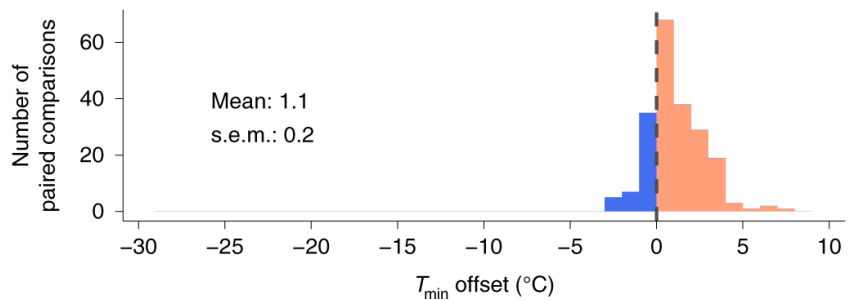
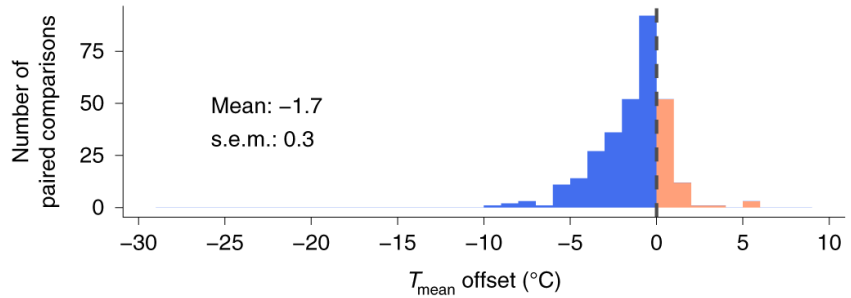
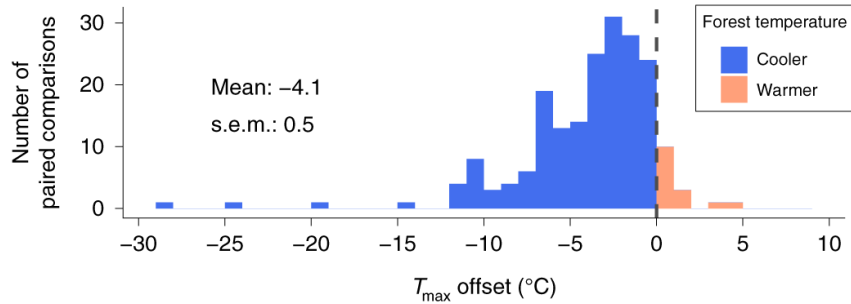


Fig. 1 Lengths of grid cells from published species distribution models (SDMs) compared to the lengths of the animals and plants they studied. See Supporting Information for methods of the literature search, the list of articles, and the data we extracted from them. Colored dots indicate the body size of a species from one study; colored horizontal lines indicate a range of body sizes if the study used multiple species. The corresponding grey dots and lines indicate the grid size (or range of sizes) of climate variables used in that study. The black density plot is a spline fitted to data from May's 1988 study [(May, 1988), fig. 6], which represents his estimate of the body size distribution of all terrestrial animals. Density plots of the rest of the terrestrial data are shown at the bottom for comparison. The two peaks in the grid-size density plots correspond to the commonly used grid scales of 1 and 10 km (e.g., these are the most common resolutions exported from the WorldClim database). We compared pairs of distributions by bootstrapping a two-sample Kolmogorov–Smirnov test (10 000 iterations), using samples ( $N = 50$ ) drawn with replacement from the observed empirical distributions. The distribution of terrestrial animal sizes was significantly shifted toward large body sizes compared to May's predicted distribution (mean  $P$ -value was 0.0028, and 99.0% of iterations had  $P < 0.05$ ). The distributions of animal and plant grid sizes were not statistically distinguishable (mean  $P$ -value = 0.31).

**a****b**

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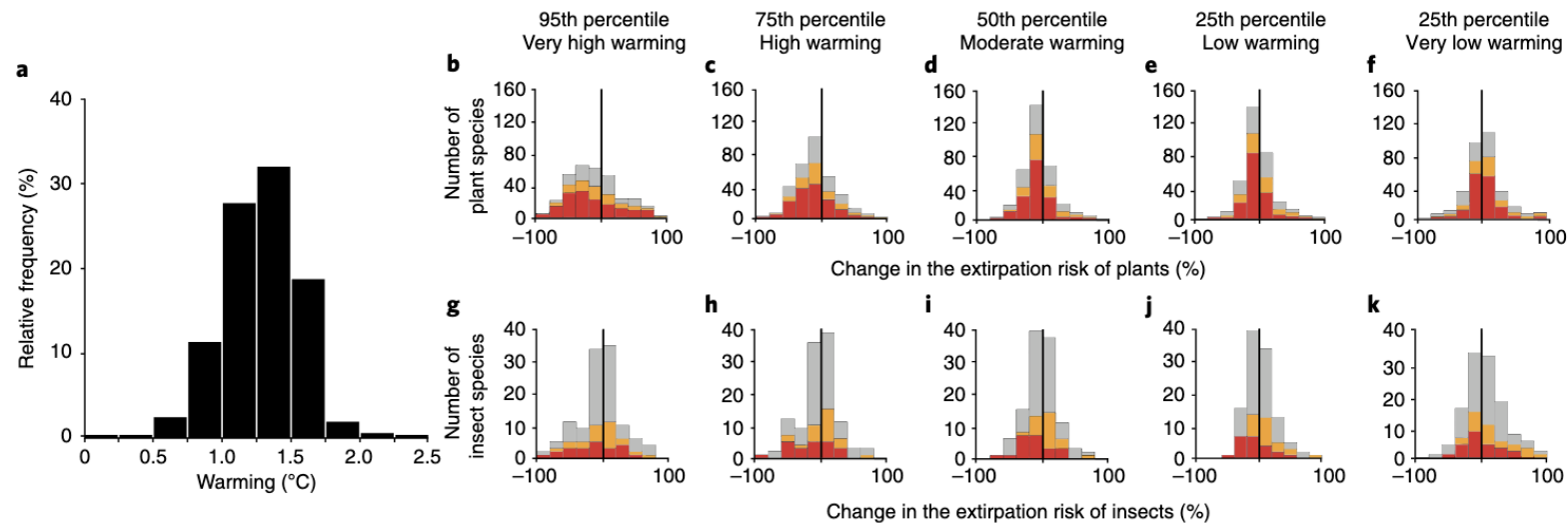
Brief Communication | Published: 01 April 2019

### Global buffering of temperatures under forest canopies

[Pieter De Frenne](#) , [Florian Zellweger](#), [Francisco Rodríguez-Sánchez](#), [Brett R. Scheffers](#), [Kristoffer Hylander](#), [Miska Luoto](#), [Mark Vellend](#), [Kris Verheyen](#) & [Jonathan Lenoir](#)

[Nature Ecology & Evolution](#) **3**, 744–749 (2019) | [Cite this article](#)

<https://www.nature.com/articles/s41559-019-0842-1>



**Fig. 2 | Modelled change in extirpation risk for each species as a function of warming and microclimatic heterogeneity.** **a**, The relative frequency of warming in each grid square. **b–k**, The modelled differences in extirpation risk between areas of high (95th percentile) microclimatic heterogeneity and low (5th percentile) microclimatic heterogeneity for various levels of warming, for plants ( $n=316$ ) (**b–f**) and insects ( $n=114$ ) (**g–k**). Red denotes species adversely affected by warming, for which the inclusion of a warming term improved model performance. Orange denotes species that exhibited a negative response to warming, but for which the inclusion of a warming term did not improve model performance. Grey denotes species that exhibited a positive response to warming (regardless of the effect of the warming term on model performance).

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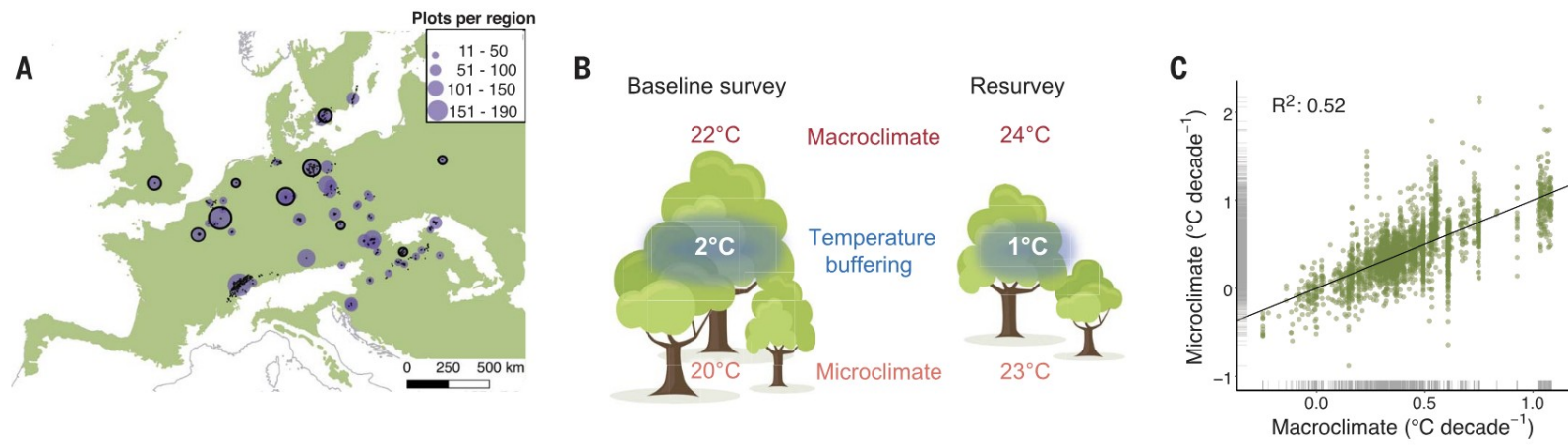
LETTERS

<https://doi.org/10.1038/s41558-018-0231-9>

## Extinction risk from climate change is reduced by microclimatic buffering

Andrew J. Suggitt<sup>1,2\*</sup>, Robert J. Wilson<sup>3\*</sup>, Nick J. B. Isaac<sup>4</sup>, Colin M. Beale<sup>2</sup>, Alistair G. Auffret<sup>5,6</sup>, Tom August<sup>4</sup>, Jonathan J. Bennie<sup>1</sup>, Humphrey Q. P. Crick<sup>7</sup>, Simon Duffield<sup>7</sup>, Richard Fox<sup>8</sup>, John J. Hopkins<sup>1</sup>, Nicholas A. Macgregor<sup>7,9</sup>, Mike D. Morecroft<sup>7</sup>, Kevin J. Walker<sup>10</sup> and Ilya M. D. Maclean<sup>1\*</sup>

<https://www.nature.com/articles/s41558-018-0231-9>



**Fig. 1. Forest microclimate change after canopy cover changes over time is considerably more variable than macroclimate change.** (A) Distribution of the 2955 resurveyed forest plots (black dots) in 56 regions (purple circles, scaled to the number of plots as indicated at the top right) across the temperate forest biome (green area) in Europe. We representatively sampled microclimate temperature in 100 forest stands, i.e., in 10 stands in each of 10 regions (black circles; effective  $n = 96$ ) to estimate the maximum (macroclimate) temperature buffering during the growing season as a function of canopy cover (23) (fig. S2). (B) Schematic overview of the method used to approximate microclimate change in the forest understory. In this example, canopy cover at the time of the baseline survey was higher than that during the resurvey, resulting in a decrease in macroclimate temperature buffering from 2 to 1°C, which in turn led to a relatively larger increase in microclimate

warming (20 to 23°C) compared with macroclimate warming (22 to 24°C). The relationship between canopy cover and the buffering of maximum macroclimate temperature was empirically assessed across the study area (fig. S2) (21). (C) Rate of macroclimate change plotted against the rate of microclimate change, with the black bisecting line representing the 1:1 relationship. Micro- and macroclimate have both significantly warmed (see text for statistical results). The distributions of values in the rates of micro- and macroclimate change are indicated by gray shading on each axis. Microclimate change rates are 45% more variable than macroclimate change rates, and macroclimate change rates only accounted for about half of the variation in microclimate change rates, as indicated by the marginal (conditional)  $R^2$  value of 0.52 (0.69). All statistical results are based on mixed-effects models with region as a random-effect (intercept) term.

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REPORT

in

## Forest microclimate dynamics drive plant responses to warming

FLORIAN ZELLWEGER , PIETER DE FRENNE , JONATHAN LENOIR , PIETER VANGANSBEKE, KRIS VERHEYEN , MARKUS BERNHARDT-RÖMERMAN

LANDER BAETEN , RADIM HÉDL , IMRE BERKI , [...], AND DAVID COOMES +24 authors [Authors Info & Affiliations](#)

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<https://www.science.org/doi/10.1126/>